Fleet Management, Fuel and Car Wash: Complete User Guide

A plain-text reference for fleet owners, managers, drivers, fuel station staff and car wash operators. Prices, taxes, permits and legal limits differ by country and change over time, so confirm them locally.

PART 1: FLEET MANAGEMENT

What it is

Fleet management is the process of buying, operating, maintaining, monitoring and finally selling a group of vehicles (trucks, buses, taxis, vans, company cars, bikes, equipment) safely, legally and at the lowest total cost. Its goals are high uptime, safety, legal compliance, low cost per km, good customer service and lower emissions.

Setting up a fleet

- List every vehicle: registration, make, model, year, chassis and engine number, fuel type, capacity, purchase date and cost.
- Store documents digitally with expiry dates: registration certificate, insurance, fitness, permit, pollution certificate, road tax.
- Create driver records: licence number, class, expiry, training, emergency contact, assigned vehicle.
- Write policies for personal use, speed, fuel, maintenance, accidents, fines and penalties.
- Install GPS tracking and, where useful, fuel sensors or fuel cards.
- Set preventive maintenance plans by km, hours or time.
- Use a spreadsheet for a few vehicles; use fleet software from about 10 vehicles.
- Track KPIs every month and improve continuously.

Buy, lease or rent

Buying gives ownership, no km limits and resale value, but needs high upfront cash and you carry repair and resale risk. Leasing gives lower upfront cost and predictable monthly payments but has km limits and exit charges. Renting is flexible and good for seasonal peaks or trials but costs most per day. Compare total cost of ownership over the holding period, including fuel, service, insurance, tax and resale. Also check service network, spare parts, warranty, safety features, efficiency and local emission rules.

Driver management

- Verify the licence with the issuing authority, check driving history and run a road test before hiring.
- Train on defensive driving, eco-driving, load securing, first aid and fatigue.
- Follow local rules on driving hours and rest; track with logs or telematics.
- Score drivers on harsh braking, acceleration, cornering, overspeed, idling and seat-belt use; reward good scores.
- Do periodic eye tests and medical checks; zero tolerance for drink or drug driving.
- Link every traffic fine to a driver using trip logs and GPS history.

Maintenance

Preventive maintenance is done at fixed km, hours or time. Predictive maintenance uses data such as fault codes and battery health. Corrective maintenance follows a breakdown. Statutory inspections are required by law. Typical preventive items (always follow the manufacturer manual): engine oil and filter every 5,000 to 15,000 km or 6 to 12 months; air filter every 10,000 to 30,000 km; fuel filter every 15,000 to 40,000 km; brakes inspected at every service and brake fluid every 2 years; tyre pressure checked weekly and rotation every 8,000 to 10,000 km; battery inspected each service and replaced in 2 to 5 years; coolant every 2 to 4 years; transmission oil every 40,000 to 100,000 km; alignment and balancing about every 10,000 km or after an impact.

Work order flow: driver reports fault, supervisor approves estimate, vehicle goes to workshop, parts and labour are recorded, quality check and road test, invoice matched to work order, history updated and next due date reset.

Daily driver check: tyres and wheel nuts, lights, indicators, horn, wipers, oil, coolant and brake fluid, brakes, mirrors, seat belt, dashboard warning lights, documents on board, safety kit (first aid, extinguisher, triangle, spare wheel) and fuel or charge level.

GPS and telematics

Telematics combines GPS, vehicle sensors and mobile data. It shows live location, routes, stops, speed, harsh events, idling, engine hours, geofence alerts, fuel level, door and temperature sensors, driver ID and engine fault codes. Benefits are faster dispatch, better ETAs, lower fuel use, theft recovery, evidence for claims and fair driver coaching. Tell employees they are tracked, state the purpose in policy, restrict data access and follow local privacy and labour law.

Compliance

Keep insurance, fitness certificate, permit, pollution certificate, road tax, driver licences and toll accounts valid at all times. Set reminders 60, 30 and 7 days before expiry. Expired documents can lead to fines, seizure, rejected insurance claims and legal liability.

Accidents

- Make everyone safe and call emergency services if needed.
- Do not move the vehicle unless required; photograph vehicles, plates, road and injuries; collect witness and other-party details.
- Report to police if the law requires, and inform the fleet manager and insurer immediately.
- Do not admit liability at the scene.
- Record cause, cost and corrective action, and share lessons with drivers.

Costs and KPIs

Fixed costs: depreciation or lease, insurance, tax, permits, interest, GPS fees, parking. Variable costs: fuel or electricity, tyres, oil, repairs, tolls, washing, distance-linked driver pay. Hidden costs: downtime, accidents, fines, idle vehicles, theft, poor routing.

- Cost per km = total operating cost / km driven.
- Utilisation % = productive days / available days x 100.
- Uptime % = (available time - downtime) / available time x 100.
- Replace a vehicle when yearly repair plus downtime cost approaches the yearly cost of a newer one.
- Track also: fuel efficiency, preventive maintenance compliance (target 95%), accidents per million km, harsh events per 100 km, on-time delivery, idle time (keep below 10%) and document compliance (100%).

Electric and alternative fuel fleets

EVs have lower energy and maintenance cost and no tailpipe emissions, but cost more to buy and need charging infrastructure. Analyse daily distance, charging windows, depot power and incentives, and start with high-use city vehicles that return to base daily. CNG and LPG can cut running cost and emissions but have fewer stations.

Choosing fleet software

Look for vehicle and driver records with expiry alerts, maintenance and work orders, fuel and GPS integration, trips and expenses, a driver mobile app, role-based access, audit trail, Excel and PDF export, local language support and fair per-vehicle pricing.

PART 2: FUEL MANAGEMENT

Why it matters

Fuel is often 30 to 40 percent of fleet operating cost, so controlling every litre gives large savings. Fuel management covers buying, storing, issuing, using and recording fuel.

Fuel types

Petrol: cars and two-wheelers; use the octane the manual recommends. Diesel: trucks, buses and SUVs; sensitive to water and microbes. CNG and LPG: lower cost and emissions; need approved cylinders and kits. Biodiesel and ethanol blends: check engine compatibility. Electricity: measured in kWh. Hydrogen: emerging with limited infrastructure.

Ways to buy fuel

Cash at the pump is simple but weak on control and open to fake receipts. Fuel cards or RFID tags tied to a vehicle give limits and automatic records. A bulk depot tank gives lower unit price but needs storage safety and theft control. Mobile fuel delivery saves driver time. Fleet apps and e-wallets give real-time control. Fuel cards are usually best for fleets: assign one card per vehicle, set limits per fill, day and month, restrict fuel type and stations, require odometer entry, block non-fuel items, change PINs when staff leave and review exceptions monthly.

Measuring consumption

- km per litre = distance / litres.
- Litres per 100 km = litres / distance x 100.
- Fuel cost per km = fuel cost / distance.
- MPG: US gallon = 3.785 litres, imperial gallon = 4.546 litres. km/l = US mpg x 0.425 = imperial mpg x 0.354.
- EV: kWh per 100 km, and cost per km = tariff per kWh x kWh per km.

Full-tank method: fill the tank fully and note the odometer. At the next full fill, note litres added. Efficiency = km since the last fill / litres added. Use the same pump, repeat several times and average. A sudden drop of more than about 10 percent needs investigation.

Fuel economy is affected by driving style, idling, load, tyre pressure and alignment, engine condition (dirty filters, bad injectors, worn plugs), route and traffic, weather, air conditioning and fuel quality.

Preventing theft and fraud

- Siphoning: locking caps, anti-siphon filters, tank sensors with drain alerts.
- Fake receipts: use cards or digital invoices with station, pump and odometer.
- Filling other vehicles or cans: alert when a fill exceeds tank capacity or the vehicle was not at the station (check GPS).
- Collusion with pump staff: rotate stations, audit pump calibration, compare with expected consumption.
- Odometer tampering: use telematics or ECU odometer data.
- Unauthorised use: trip logs, geofences, after-hours alerts.

Sensors and bulk storage

Fuel-level sensors show refuel and drain events. Flow meters measure engine consumption. CAN bus and OBD data read use from the engine computer. RFID pump automation records litres, vehicle and odometer. Tank gauging probes report stock, water and leaks.

For bulk tanks follow local licensing, fire and environmental rules. Use approved tanks with secondary containment, earthing, no smoking or ignition sources, extinguishers and sand, trained staff and spill kits. Drain water from diesel tanks regularly. Reconcile daily: opening stock + deliveries - issues = expected closing stock; compare with the physical reading. Variance under about 0.5 percent is usually normal; repeated or large negative variance suggests leaks, theft or meter error.

Saving fuel

- Cut idling: switch off if stopping over about a minute where safe.
- Drive smoothly and at moderate speeds; use cruise control on open highways.
- Check tyre pressure weekly and service on time.
- Remove extra weight and roof racks.
- Plan routes, combine loads and avoid empty returns.
- Use the right vehicle size for each job.
- Reward the most efficient drivers and use trusted stations with fleet rebates.

EV charging

AC chargers (about 3 to 22 kW) suit overnight depot charging; DC fast chargers (50 kW and above) suit quick top-ups. Energy needed = distance x kWh per km, plus a 15 to 20 percent margin. Charging time is roughly energy needed / charger power, slower above about 80 percent. Charge off-peak, stagger vehicles to avoid overload, avoid routinely charging to 100 percent or deep discharge, and track kWh and cost per vehicle.

PART 3: CAR WASH

Types of car wash

- Hand wash: gentle and detailed, but labour heavy.
- Self-service bay: low staff cost, quality depends on the customer.
- In-bay automatic: compact, one operator, lower throughput.
- Tunnel wash: high volume and consistent, high investment.
- Touchless: no brushes so lowest scratch risk, may leave film on heavy dirt.
- Soft-cloth: good cleaning if cloth is clean and well maintained.
- Mobile or doorstep wash: convenient, low rent, needs water access.
- Waterless or rinseless: saves water, not for heavy mud.
- Steam wash: hygienic, low water, slower.

Exterior process

- Inspect with the customer, note existing damage and remove valuables.
- Pre-rinse to remove loose sand; never rub dry dirt.
- Clean wheels and tyres first with separate brushes.
- Apply foam with a pH-neutral shampoo and let it dwell, out of direct sun.
- Wash with a clean mitt from top to bottom, using the two-bucket method.
- Rinse thoroughly, then dry with clean microfibre, a chamois or a blower.
- Finish with tyre dressing, glass cleaner and optional wax, then inspect with the customer.

Interior process

Remove mats and personal items. Vacuum seats, carpets, boot and crevices. Wipe dashboard and panels with a suitable cleaner and avoid greasy shine on the steering wheel and pedals. Clean glass with a streak-free cleaner. Treat fabric stains and condition leather. Find the cause of odours such as spills or damp.

Extra services

Spray wax lasts 1 to 3 months, sealant 4 to 6 months and ceramic coating several years with proper care. Other add-ons: clay bar decontamination, machine polishing, engine bay cleaning, underbody wash after mud or salt, headlight restoration, steam interior deep clean, odour removal, leather care and tyre shine.

Chemicals, water and environment

- Use pH-neutral car shampoo; dishwashing liquid strips wax and dries rubber.
- Use acid-free wheel cleaner on most alloys; use degreaser for engine bays and rinse well.
- Use ammonia-free glass cleaner on tinted film.
- Keep safety data sheets, label containers, dilute exactly as directed, wear gloves and eye protection, and never mix chemicals.
- Wash microfibre separately without fabric softener.
- Hand washing with a hose can use 100 to 300 litres; pressure washers and recycling systems cut this a lot. Recycling systems often reuse 60 to 90 percent of water.
- Drain bays to a sediment trap and oil-water separator, never straight into storm drains.
- Use softened or RO water for a spot-free final rinse.
- Check local licences for water use and effluent discharge.

Equipment

Pressure washer, foam lance, air compressor, wet and dry vacuum, water tank and pump with softener or RO unit, lifts or ramps, blowers, good lighting, billing system, CCTV and a comfortable waiting area. Keep spare nozzles, seals, filters and brushes to avoid downtime.

Pricing and business

Calculate cost per wash (water, power, chemicals, labour, rent, depreciation, payment fees, licences), add a profit margin and compare local rates for each vehicle size. Offer packages: Basic (exterior), Standard (exterior plus interior vacuum and glass), Premium (adds wax, door jambs, deep mats) and Detailing (adds clay, polish, sealant

or coating). Add memberships, fleet contracts with monthly invoicing and off-peak discounts. Break-even washes per month = fixed monthly costs / (average price - average variable cost).

To start: study traffic and competitors, choose the format, secure a site with water, power and drainage, get licences, budget for equipment and working capital, train staff, set up billing and liability insurance, and market with launch offers, a maps listing and referral rewards. A low-cost start is a mobile or small hand-wash setup.

Safety and customer care

- Photograph pre-existing damage and get customer acknowledgement.
- Use slip-resistant footwear, gloves, eye protection and wet-floor signs.
- Use waterproof sockets, residual-current devices and earthing.
- Control keys, never leave vehicles running, return any valuables found.
- Have a quick complaint and damage procedure and carry insurance for vehicles and workers.

Washing fleets

Taxis and delivery vans may need washing daily or every other day; trucks weekly. Wash at the depot after return or before dispatch, log date and cost per vehicle, keep cameras, sensors, number plates and lights clean, sanitise bus and school vehicle interiors, and use rinseless wash where there is no drainage.

PART 4: INTEGRATING FLEET, FUEL AND CAR WASH

Link all records to one vehicle timeline: trips, fuel fills, washes, repairs, documents and costs. Before a trip do the inspection, assign the driver, check fuel or charge and clean the vehicle if needed. During the trip use GPS, refuel only at approved stations and record odometer. After the trip close the trip, log defects, reconcile litres against distance and wash on schedule. Monthly, review cost per km against budget, fuel efficiency and exceptions, maintenance compliance, documents expiring in 60 days, accidents and violations, wash cost and quality, and set action items with owners and due dates.

Daily checklist: vehicles and drivers confirmed, inspections done, fuel issued and odometer captured, GPS online, breakdowns handled, vehicles parked and keys logged, entries completed. Weekly: tyres and fluids, driver alerts, fuel exceptions, upcoming services and renewals. Monthly: reconcile fuel cards and tolls, check tank variance, audit assets. Yearly: insurance, fitness and permit renewals, replacement plan, driver refresher and medicals, policy review, car wash equipment audit.

Emissions estimate: litres x about 2.3 kg CO2 for petrol or about 2.7 kg for diesel (approximate).

PART 5: FREQUENTLY ASKED QUESTIONS

Q. What is the difference between fleet management and a GPS tracker?

A tracker only shows location and basic events. Fleet management covers vehicles, drivers, maintenance, fuel, compliance, costs and safety. GPS is one tool within it.

Q. How do I reduce fleet costs quickly?

Start with fuel: stop idling, control cards, coach drivers and plan routes. Then improve preventive maintenance and tyre care, remove under-used vehicles and renegotiate insurance and vendor rates.

Q. When should I replace a vehicle?

When repair and downtime cost exceed the cost of a newer vehicle, when safety or emission rules are not met, or when resale value starts falling sharply. Use cost per km by vehicle to decide.

Q. How do I stop personal use of company vehicles?

Set a written policy, use GPS geofences and after-hours alerts, require trip logs and apply fair consequences. If personal use is allowed, define limits and record it separately.

Q. How often should vehicles be serviced?

Follow the manual, using the shorter interval for severe duty such as heavy loads, dust, stop-start traffic or heat. Use km, hours or time, whichever comes first.

Q. What if insurance expires?

The vehicle may be illegal to drive, fines can apply and claims can be rejected, leaving you liable. Use reminders and renew before expiry.

Q. How do I handle traffic fines?

Match the notice time and vehicle with trip and GPS logs, inform the driver, and pay or recover as per policy before extra penalties apply.

Q. Are EVs good for fleets?

Yes for predictable daily routes with depot charging, where low energy and maintenance cost offset the higher price. Check range, payload, charging and incentives, and trial a few vehicles first.

Q. Why is my fuel consumption suddenly higher?

Common causes: low tyre pressure, heavy load, harsh driving, more idling, traffic, dirty filters, engine faults, misalignment, air conditioning, bad fuel, or theft and leaks. Inspect the vehicle and compare trip data.

Q. How do I detect fuel theft?

Use tank sensors with drain alerts, compare fills with tank capacity and GPS location, compare consumption with baseline, secure caps, use limited fuel cards and audit bulk tank variance.

Q. Are fuel cards better than cash?

Usually yes for fleets: limits, automatic records, odometer prompts, lower fraud and easier reconciliation. Cash may suit very small operations.

Q. Should I use premium fuel?

Only if the manual recommends it. Otherwise it is an unnecessary cost.

Q. How much does idling waste?

A car may burn roughly 0.5 to 1 litre per hour idling and large diesel engines more. Across a fleet this is a large avoidable cost.

Q. AC or windows open?

At low speed, windows open usually uses less fuel. At highway speed, open windows add drag so AC is often similar or better.

Q. How can I prevent water in diesel?

Buy from reputable suppliers, keep tanks full, drain water regularly, use water-separator filters, change filters on schedule and keep storage clean and sealed.

Q. How can an EV fleet cut charging cost?

Charge at the depot off-peak, avoid peak demand with smart chargers, track kWh per vehicle and compare cost per km with diesel or petrol.

Q. What are the fuel station safety rules?

Switch off the engine, no smoking or flames, follow phone rules at the dispenser, do not overfill, and report spills immediately.

Q. How often should I wash my car?

About every 2 weeks, weekly in dusty, coastal or salty conditions, and promptly after bird droppings, sap or insects, which can damage paint.

Q. Touchless or brush wash?

Touchless avoids brush contact but may leave film on heavy dirt. Well-maintained soft-cloth cleans better. Hand washing with clean tools is gentlest.

Q. Can automatic washes scratch paint?

Old, dirty or abrasive brushes can cause swirl marks. Well-maintained soft-cloth or touchless systems are generally safe.

Q. Wax, sealant or ceramic coating?

Wax shines for weeks to months, sealant lasts longer, ceramic forms a hard layer lasting years but costs more and needs correct preparation.

Q. How long does a wash take?

A basic exterior hand wash takes 20 to 40 minutes, interior adds 20 to 40 minutes, detailing takes hours, and tunnels take a few minutes.

Q. How do I remove bird droppings or sap?

Soften with a wet microfibre or quick detailer and wipe gently. Use a sap remover for stubborn sap. Never scrub dry material.

Q. How do I remove bad smell in the car?

Remove the source, vacuum, clean upholstery, replace the cabin filter if needed, dry fully and use an odour neutraliser. Severe cases may need ozone treatment.

Q. Is waterless wash safe?

Yes for light to moderate dust with enough product and clean microfibre. Rinse heavy mud and sand first.

Q. Can I wash an EV normally?

Yes, but avoid high-pressure water at the charging port, battery seals and sensors, close the port and follow the manufacturer guidance.

Q. Is it safe to wash the engine bay?

With care: cover the alternator, air intake and fuse box, use low pressure and a suitable degreaser, rinse gently, let it dry and never do it on a hot engine.

Q. How do I fix hard water spots?

Dry quickly, avoid sun washing, use RO or softened water for the final rinse, remove existing spots with a mild remover and protect the paint.

Q. How do I get repeat car wash customers?

Give consistent quality, keep the site clean, offer loyalty points or memberships, accept digital payments, send reminders and resolve complaints fast.

Q. What should I track first if I have no system?

Odometer readings, fuel fills and maintenance dates for every vehicle. These quickly give cost per km and reliability data.

Q. Can this guide replace legal advice?

No. Taxes, permits, safety standards, driving hours, privacy and environmental laws vary, so confirm with local authorities or qualified professionals.